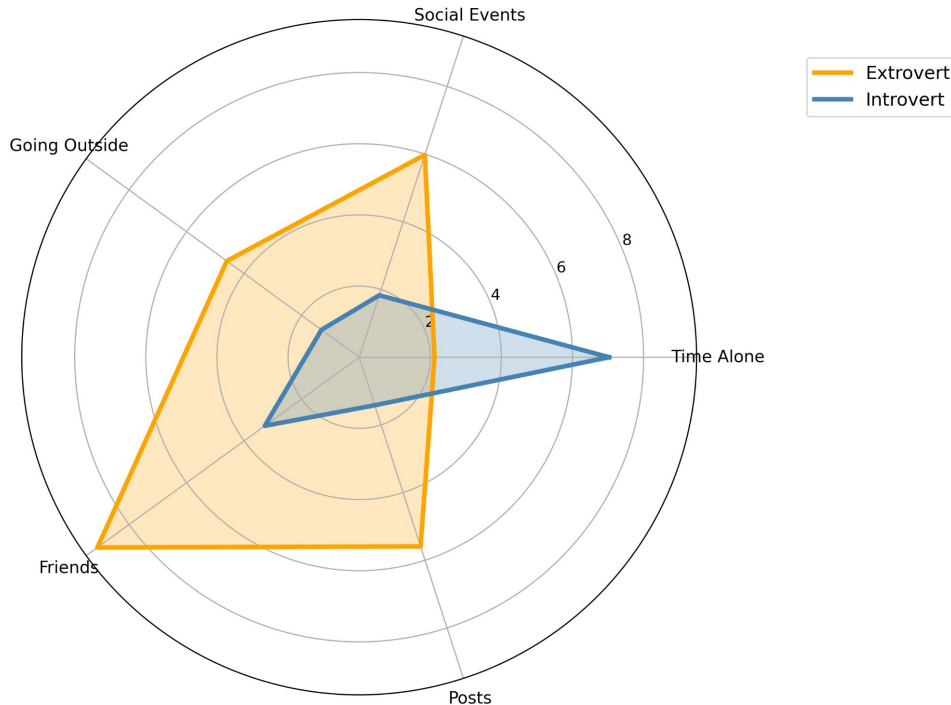


Social Batteries: Introverts vs. Extroverts



Student Information

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Exploring behavioral profiles across social events, solitude, and digital engagement to define the personality spectrum.

1. Extrovert vs. Introvert Behavior Data

Dataset: personality_dataset: Explore and Predict Social Behaviors and Personality Types

URL: <https://www.kaggle.com/datasets/rakeshkapilavai/extrovert-vs-introvert-behavior-data>

Dataset Overview

The dataset contains behavioral and social information for **2,900 individuals**, classified as introverts or extroverts. It includes variables related to social interaction, personal habits, and online activity, making it suitable for analyzing personality traits and identifying behavioral differences between the two groups.

Key Features

- **Time_spent_Alone:** Daily hours spent alone (0-11).
- **Stage_fear:** Presence of stage fright (Yes/No).
- **Social_event_attendance:** Frequency of social events (0-10).
- **Going_outside:** Frequency of going outdoors (0-7).
- **Drained_after_socializing:** Feeling drained after socializing (Yes/No).
- **Friends_circle_size:** Number of close friends (0-15).
- **Post_frequency:** Social media posting frequency (0-10).
- **Personality:** Target variable (Extrovert/Introvert).

Research Question

PRIMARY QUESTION

Which social behaviors most strongly distinguish introverts from extroverts?

SECONDARY INVESTIGATIONS



Do introverts spend more time alone?



Do extroverts attend more social events?



Which behavior shows the largest difference?

About Data

The dataset contains **2,900 observations**, well-balanced between **1,491 Extroverts** and **1,409 Introverts**.

There are **8 columns** and no missing values were detected, even if the dataset documentation mentions missing values; however, exploratory analysis of the downloaded version revealed no missing observations.

5 NUMERICAL VARIABLES

- **Time_spent_Alone:** Daily hours spent alone (0-11).
- **Social_event_attendance:** Frequency of social events (0-10).
- **Going_outside:** Frequency of going outdoors (0-7).
- **Friends_circle_size:** Number of close friends (0-15).
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3 CATEGORICAL VARIABLES

- **Stage_fear:** Presence of stage fright (Yes/No).
- **Drained_after_socializing:** Feeling drained after socializing (Yes/No).
- **Personality:** Target variable (Extrovert/Introvert).

Methodology

DATA VARIABLES

Numerical (Ratio): Variables measuring social behaviors such as time spent alone, social event attendance, friendship network size, and social media activity.

Categorical (Nominal): Personality traits like stage fear and social exhaustion.

INSTRUMENTS

Environment: Python via Jupyter Notebook.

Analysis: Pandas for exploration and descriptive statistics.

Visualization: Matplotlib and Datawrapper.

DATA QUALITY CHECKS

Verified variable types, class distributions, and missing values.

ANALYSIS FOCUS

Focused on descriptive statistics and group comparisons (Introverts vs. Extroverts).

- Bar charts for categorical comparisons via Datawrapper.
- Box plots for distribution analysis via Python/Matplotlib.

Reproducibility

DEVELOPMENT ENVIRONMENT

The analysis was conducted using **Python** within a **Jupyter Notebook** environment.

MAIN LIBRARIES

- Python 3.13
- Pandas 2.3.3
- Matplotlib 3.10.6
- NumPy 2.3.5

REPRODUCTION REQUIREMENTS

All data preparation, statistical analysis, and visualizations can be reproduced using the public Kaggle dataset, the attached Jupyter Notebook, and the GitHub repository.

GITHUB REPOSITORY



Source code and notebook available at:

<https://github.com/tibalbi/Social-Batteries-Introverts-vs.-Extroverts>

Data Limitations & Potential Biases

DATA COLLECTION

Self-Reporting: Subject to response bias and inaccurate self-perception.

Metadata Gap: Limited info on recruitment, demographics, and procedures.

VALIDITY & LABELS

Variable Interpretation: Terms like "social events" may vary by individual.

Label Origins: Methodology for personality labels (real vs. synthetic) is unclear.

SAMPLE REPRESENTATIVENESS

The representativeness of the sample cannot be fully assessed due to the lack of exhaustive demographic metadata.

DATA ENRICHMENT

The dataset could be enriched with demographic variables such as age, gender, and occupation to enable a more comprehensive analysis of the factors associated with introversion and extroversion.

INTERPRETATION MANDATE

Findings should be interpreted strictly as **descriptive patterns** within this specific dataset and should **not be generalized** to the broader population.

Data Analysis Results

TIME ALONE

Introverts spend much more time alone than extroverts: on average 7.03 hours versus 2.12.

SOCIAL EVENTS

Extroverts participate in events more frequently: on average 5.98 events versus 1.83 for introverts.

FRIENDSHIP NETWORK

Extroverts boast larger social networks, with an average of 9.10 friends compared to 3.28 for introverts.

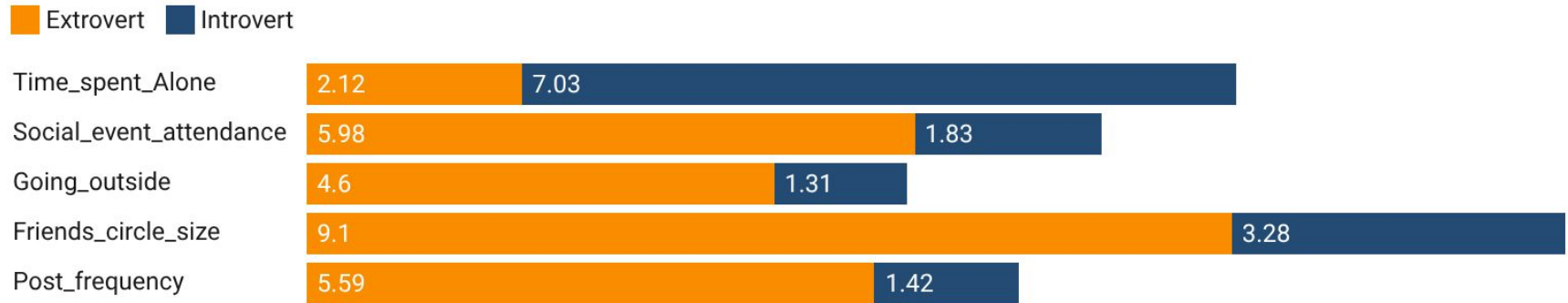
SOCIAL MEDIA

The social media posting frequency of extroverts is nearly four times higher than that of introverts.

Overall, the results indicate that **social engagement behaviors** strongly differentiate introverts from extroverts. Time spent alone, **friendship network size**, and **social event attendance** emerged as the most informative variables.

Social Engagement Behaviors: A Comparison Between Introverts and Extroverts

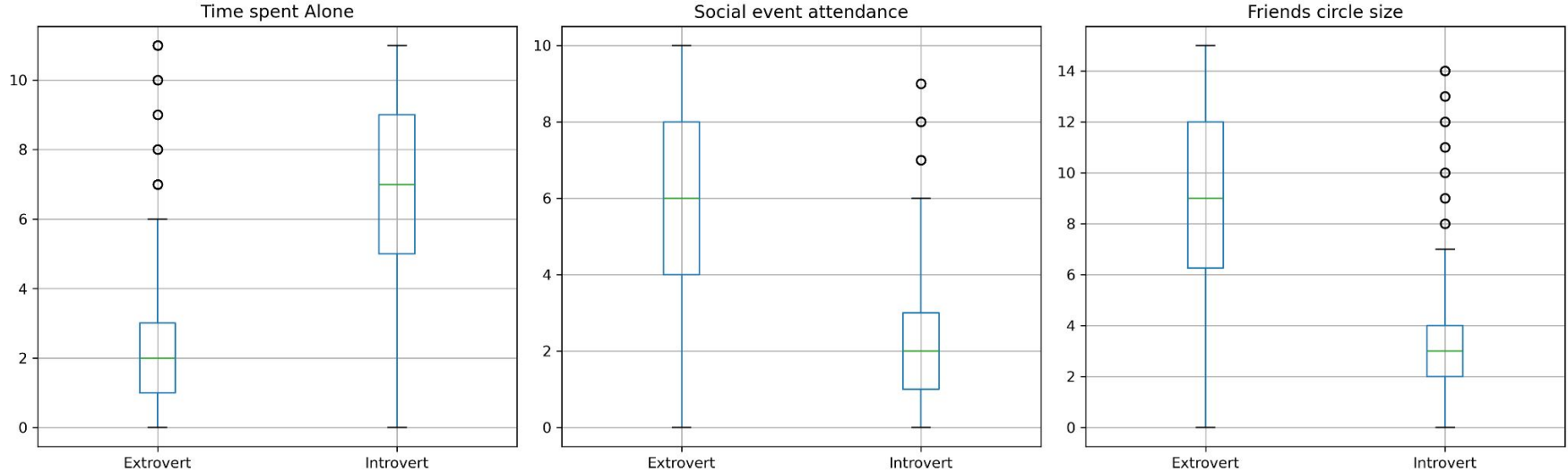
[Introverts vs Extroverts: Average Behavioral Characteristics]



Created with Datawrapper

Distribution of Behavioral Variables by Personality Type

Behavioral Differences Between Personality Types



KEY INSIGHT

Social circle size, time spent alone, and social event attendance show the greatest separation between introverts and extroverts, making them the most informative behavioral indicators in the dataset. The limited overlap between the distributions confirms that these variables provide strong separation between personality types.

LICENCE

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Dataset source:

Extrovert vs. Introvert Behavior Data (Kaggle)

Dataset license:

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